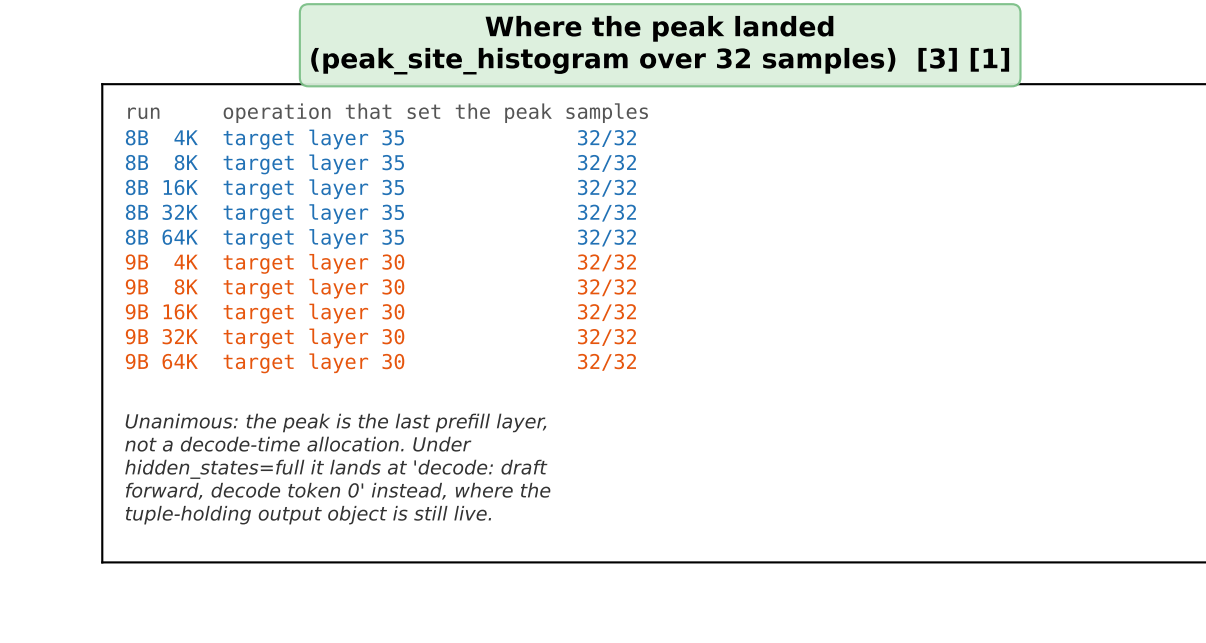
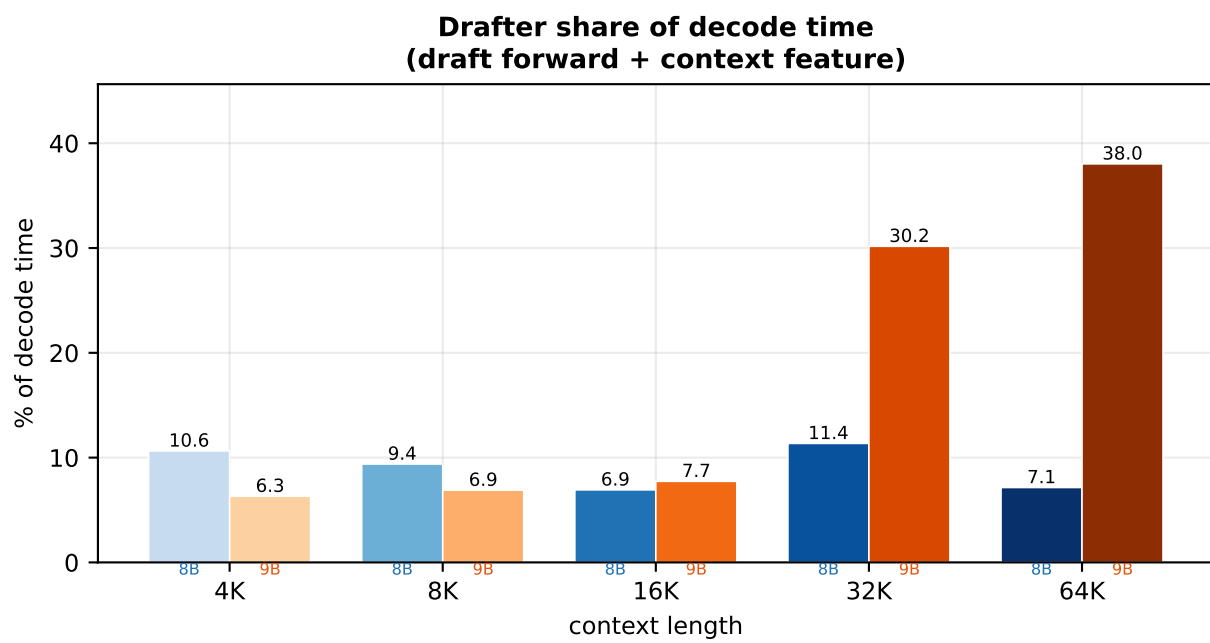
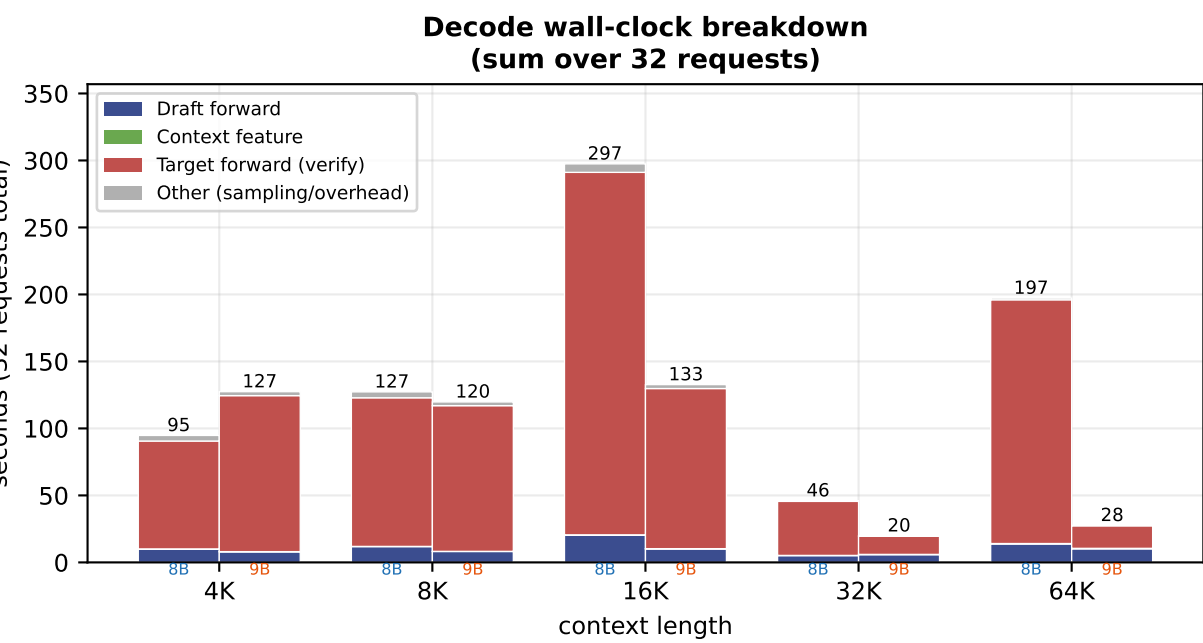
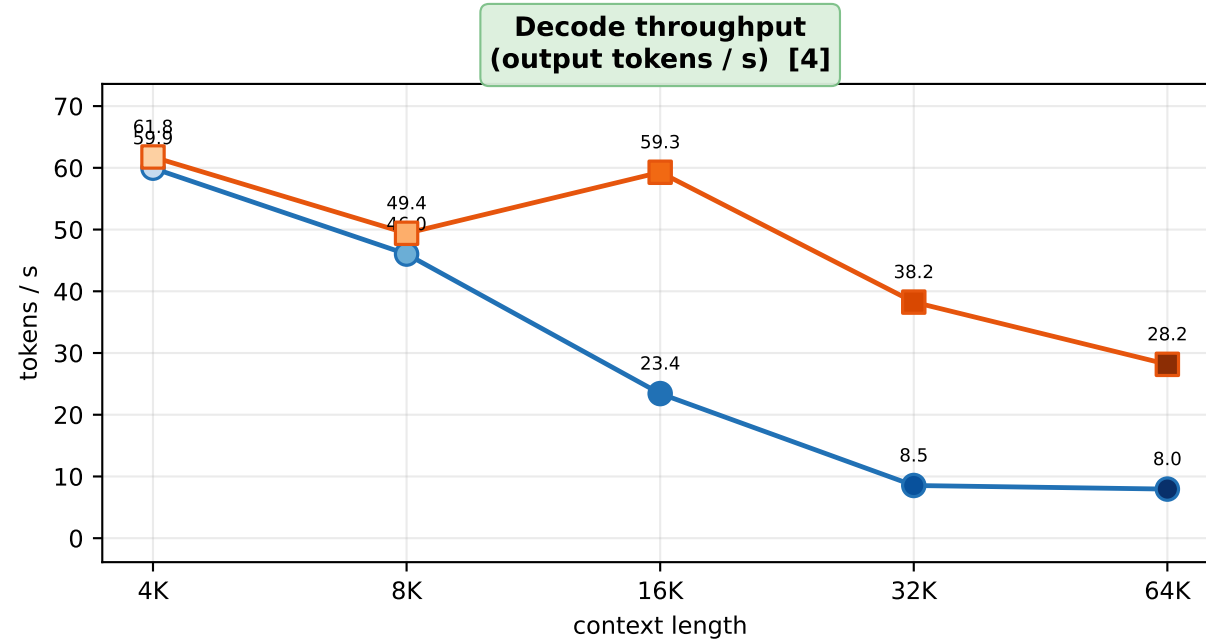
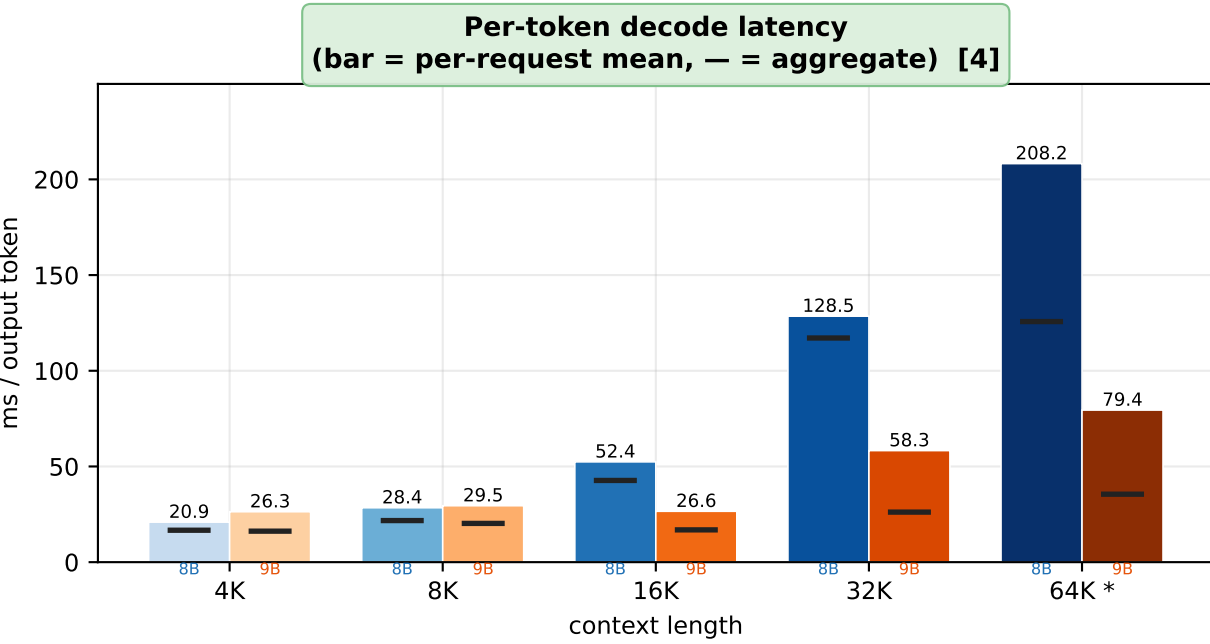
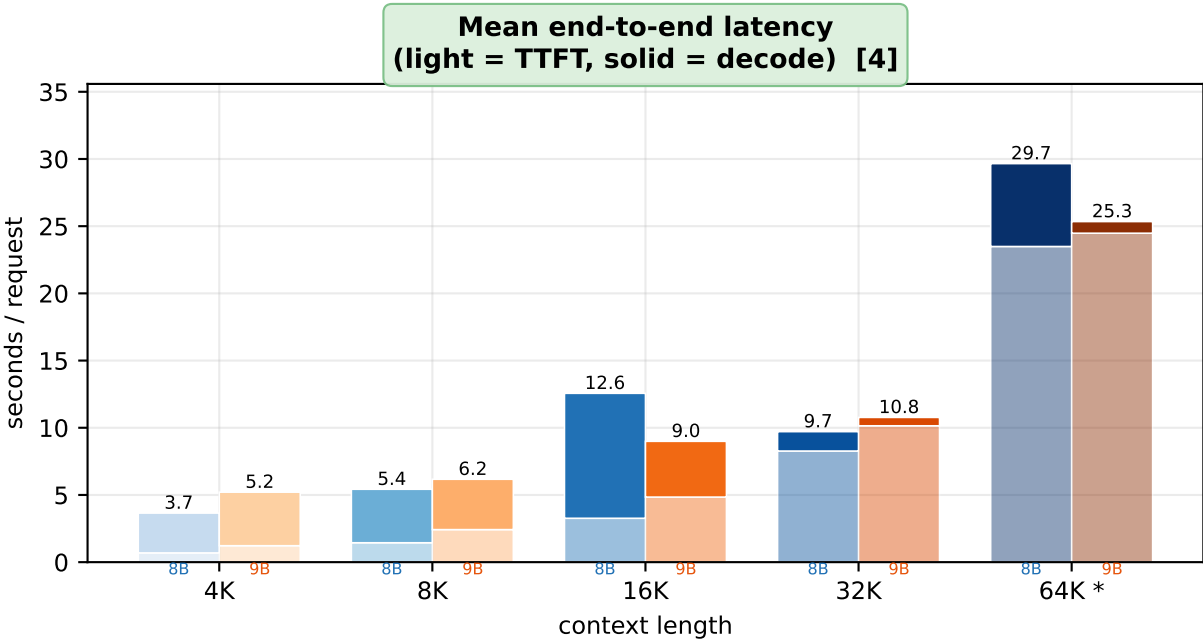
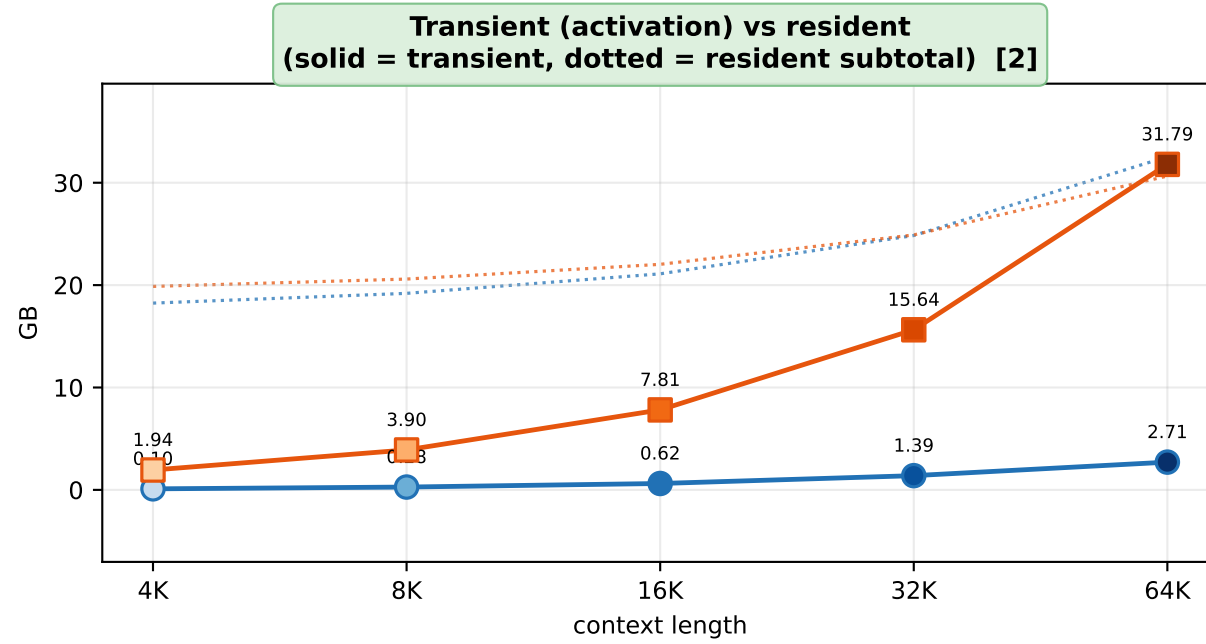
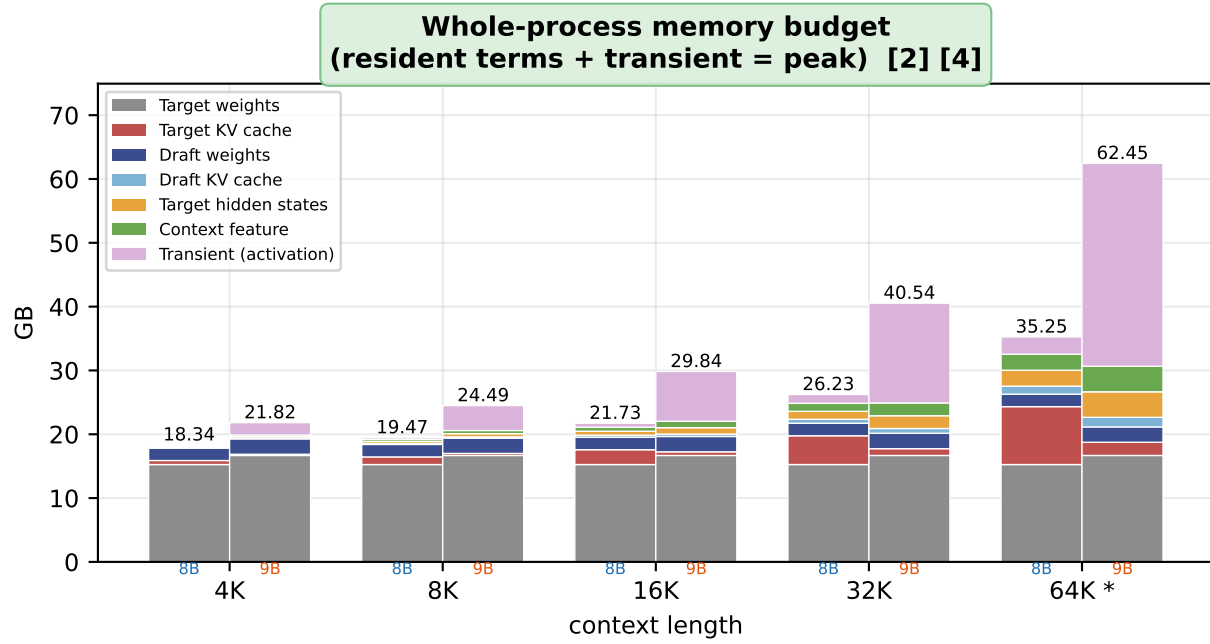
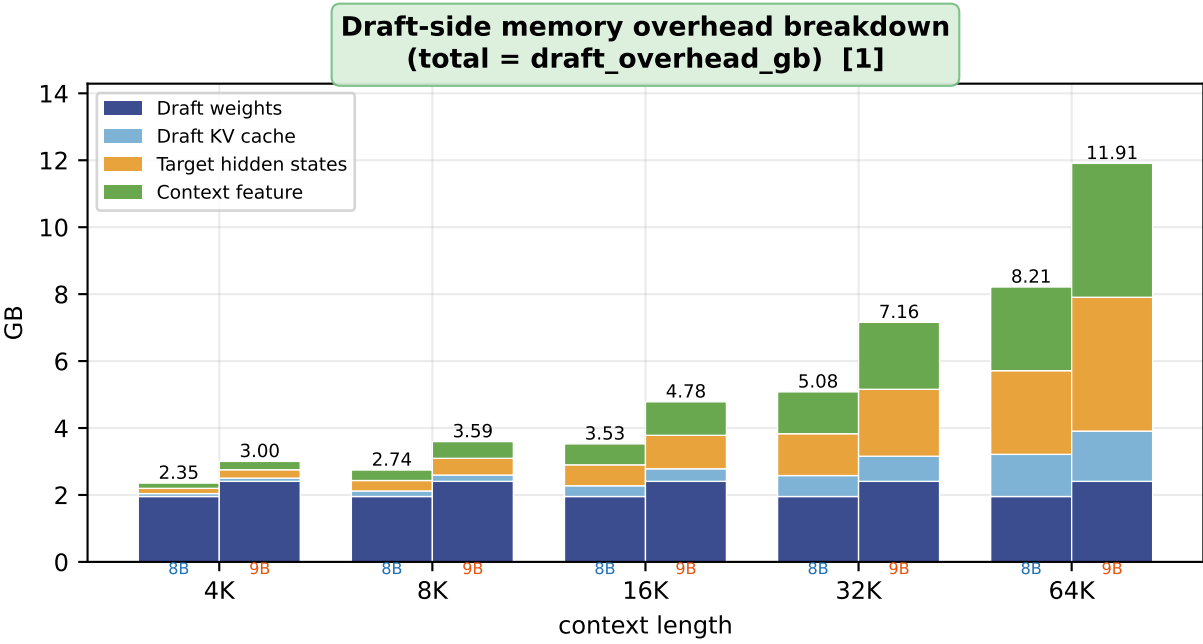
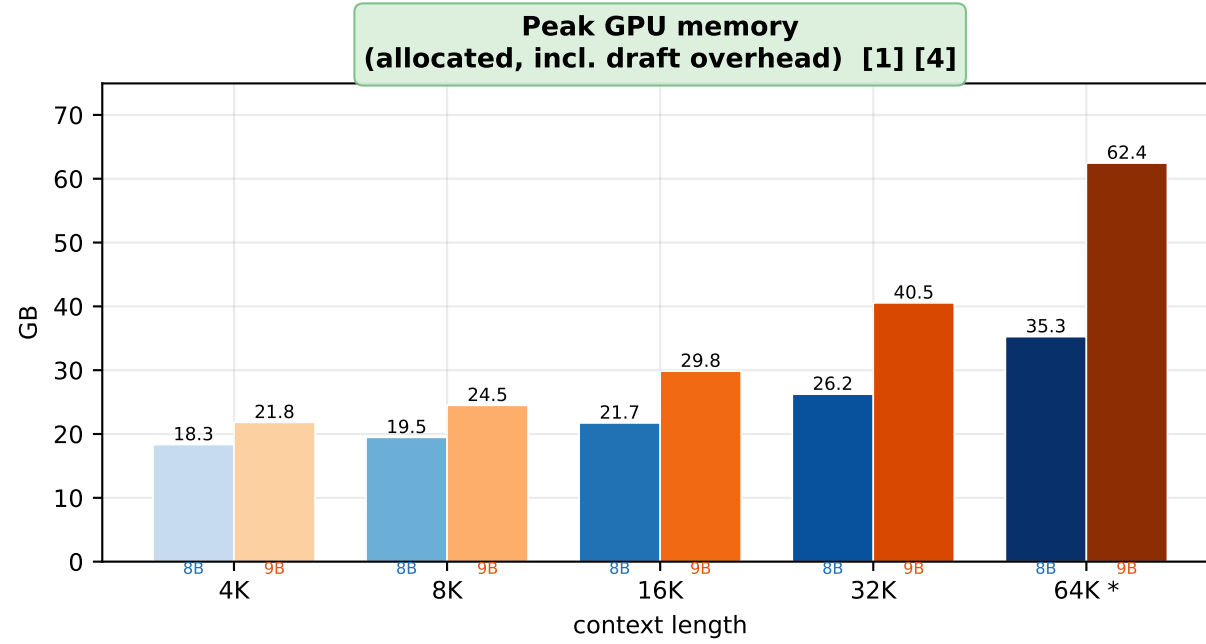
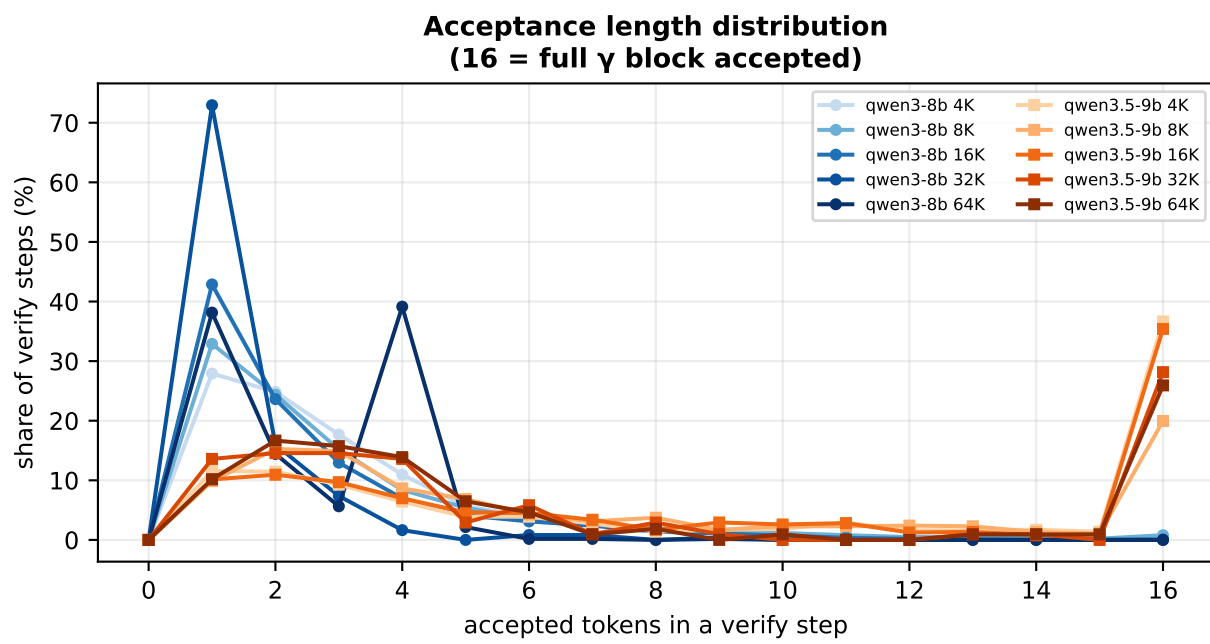
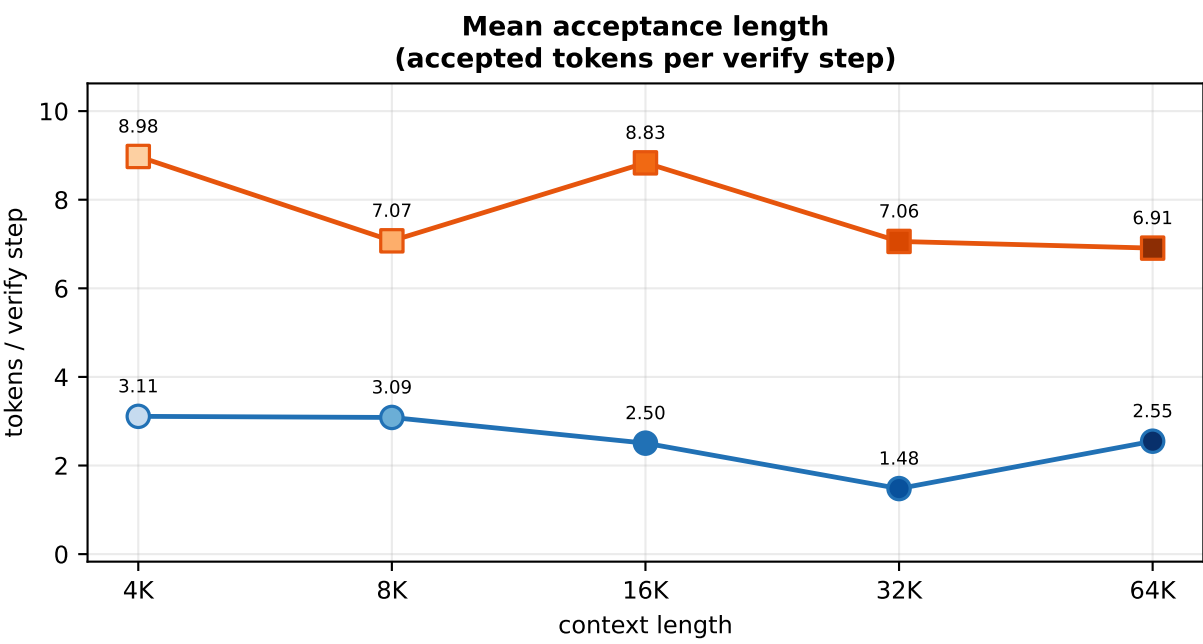


DFlash speculative decoding — LongBench-E benchmark-wide averages, hidden_states=selective (Qwen3-8B vs Qwen3.5-9B @ 4K / 8K / 16K / 32K / 64K, block=16, gamma=15, greedy)



Qwen3-8B · 4K Qwen3-8B · 8K Qwen3-8B · 16K Qwen3-8B · 32K Qwen3-8B · 64K Qwen3.5-9B · 4K Qwen3.5-9B · 8K Qwen3.5-9B · 16K Qwen3.5-9B · 32K Qwen3.5-9B · 64K Qwen3-8B (trend) Qwen3.5-9B (trend)

Samples per context length [8B / 9B where they differ] — 4K: n=32 (13 datasets) · 8K: n=32 (12 datasets) · 16K: n=32 (10 / 9 datasets) · 32K: n=32 (9 datasets, 31 composed) · 64K: n=32 (8 datasets, 32 composed)

Mean output tokens per request [8B / 9B] — 4K: 178 / 246 · 8K: 183 / 185 · 16K: 218 / 246 · 32K: 12 / 24 · 64K: 49 / 24 (long-context runs emit far fewer tokens; compare per-token metrics, not per-request ones)

Panels highlighted in green differ from visualization/ (hidden_states=full) — ADDED = a quantity that did not exist in those records, REVISED = same quantity, changed measurement

- [1] REVISED hidden_states=selective. The target's residual streams reach the drafter through forward hooks on the injected layers (HiddenStateTap) instead of output_hidden_states=True, so max_target_hidden_states_gb counts n_inj layers (5 on 8B, 8 on 9B) rather than L_t+1 (37, 33). The context feature is bit-identical: mean acceptance length and mean output tokens match the full-mode records at all 10 points.
- [2] ADDED summary*.memory_budget, a new record field. Splits the peak into resident terms and the remainder (transient_gb = peak - resident). Needs target_weight_gb = module_bytes(target), also new, which the full-mode records do not carry -- they accounted for the drafter's share only.
- [3] REVISED peak_site now stamps the decode position an interval *opened* at, not the one current when it closed. Previously an interval that opened at the last prefill layer and was still open when decoding began was reported as 'target layer N, decode token 0'. A boundary is also closed at the end of prefill for every configuration, not only block_size>1. peak_site_histogram (added) counts the setting operation per sample.
- [4] REVISED Device count differs from the full-mode records: 8B/64K 2->1, 9B/32K 2->1, 9B/64K 3->2. Selective frees enough memory to fit on fewer cards. 9B/64K is still sharded and starred on the x axis, so its timings are pipeline-parallel. Memory stays comparable across a device change; timings do not -- going from sharded to single-GPU speeds the *baseline* up more than it speeds DFlash, so a speedup can read lower without DFlash having got slower.